

Datathon 2026 Submission Report

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Track 1(Insights) & Track 2 (Predictive)

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Problem

To tackle the problem of figuring out which school districts might need educational changes in the future, this project focuses on creating predictive models to spot early on which areas need extra academic help. By predicting whether students in a certain district might be at risk, these models help plan how to share resources and assign teachers more effectively. In the end, this method gives a clearer evaluation of student performance, letting policymakers step in sooner and create a fairer, more effective learning environment.

Approach

In this project, we start step by step by following the technical processes, beginning with systematically collecting data from the relevant government sources. Then, the data is cleaned and combined into a special CSV file. This basic data can be merged with the secondary master dataset to create foundational demographic information. The model focuses on the most impactful features by doing feature selection on numerical data using ANOVA and checking categorical features with the Chi-square test. The predictive framework is built by training and testing lots of machine learning models like Logistic Regression, Decision Trees, and Random Forest, and for Random Forest, we also do hyperparameter tuning to keep its performance at the best level.

We use cross-validation methods to test and make sure the model's reliability and general usefulness for future scenarios, and we also test it with data from 2025 that the model hasn't seen at all yet.

Key Insights

Based on the goals, processes, and strategies, the main points regarding this project are as follows: Shifting from a reactive model to proactive interventions: The key point is transitioning from a traditional trial-and-error approach to strategies that can be implemented in practice as early as possible. By identifying and addressing problematic areas, schooling plans can be set up before success decreases, and actions can be carried out based on 'prevention' rather than a low level of 'correction' awareness. Effectiveness in using data-based resources: This approach provides a scientific basis for distributing resources efficiently. Without allocating arbitrarily, authorities can use evidence-based estimates to determine where teachers or special facilities are needed in each region and allocate resources and teachers accordingly.

Feature Analysis Using Combination: In the study, we made sure to balance and emphasize both quantitative measures of improvement (like exam scores) and non-quantitative data (like zoning conditions or types of teaching) by using ANOVA (for numerical data) and Chi-square (for categorical data). By selecting features at these two levels, we can better understand educational success in depth.

Unexpected Data and Future Adjustments: Using 2025 data as unseen data is important because it allows us to test how well the model can perform with future educational scenarios.

This ensures that the model can realistically predict future educational needs without just memorizing past trends (overfitting).

Improving Algorithm Confidence with Ensemble Learning: Simple models like Logistic Regression provide the basic foundational data, while combining Random Forest with Hyperparameter Tuning in this project shows a particular focus on accuracy and the ability to handle complex, non-linear relationships within educational data. This means district evaluations can be relied on confidently.

Integrated Data Ecosystem: Combining primary and secondary datasets into a single master CSV shows that educational success cannot be fully seen just by looking at one dataset. By merging different data sources, the model can better capture various socio-economic and organizational factors that impact district performance.